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PDF Accessibility Project

Problem Context and Project Summary

Many instructors struggle with complying with ADA guidelines, especially when uploading PDFs to their courses online. Although there are many public accessibility checkers online, many require fees and subscriptions to their services, and may not fully work on PDFs. Most screen readers and other public accessibility checkers fail in capturing the Alternative Text criterion, for example, because the text does not give the correct context to its associated graphic. Therefore, a Multi-head Transformer Model will be implemented to help in identifying PDF layouts and will be combined with a CNN architecture to pinpoint specific WCAG 2.1 AA criterion while focusing on the semantic meaning within Alternative Text. The other criteria this project focuses on are Color and Contrast and Table Structure. The model being used in this project is the LayoutMv3 model, published from HuggingFace. The model will go through the first dataset, DocLayNet, to identify PDF elements, such as headings, tables, and images. Afterwards, the model will go through a second dataset PDF Accessibility Benchmark (2025), with data augmentation to increase the dataset size, to understand where to find failures in WCAG 2.1 AA criteria. The model will analyze PDFs as input and output confidence scores for each criterion, thus helping the user identify which criteria to focus on. Along with the confidence scores, an overall accessibility score will also be published. Weights for each criterion, with the Alternative Text being the highest, will be added into the accessibility score equation.

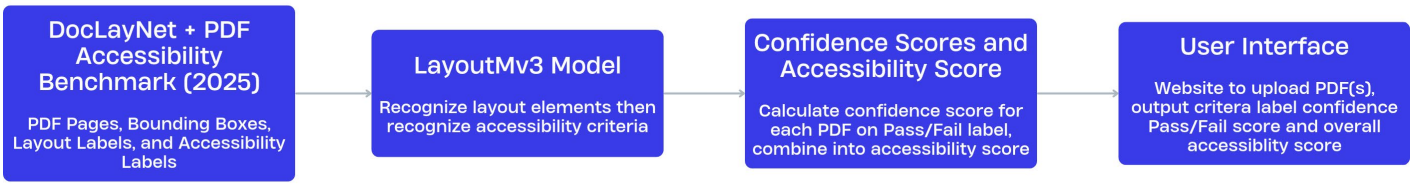
Dataset

The two datasets used in this project is the DocLayNet dataset and the PDF Accessibility Benchmark (2025) dataset. The DocLayNet dataset is accessible through the HuggingFace datasets library and contains more than 80,000 PDF pages that were manually annotated using several of the 11 provided PDF layout classes (Caption, Footnote, Formula, List-item, Page-footer, Page-header, Picture, Section-header, Table, Text, Title). Each PDF is associated with one of the 6 categories (Financial reports, Scientific articles, Laws and regulations, Government tenders, Manuals, and Patents). Because this dataset is very large and contains lots of unnecessary data, we will need to create a subset that is only relevant to what this project needs (for example, we do not need the category names or any other metadata for each PDF). There also seems to be a class imbalance within the layout classes (category_id 10 has the highest count, which is the Text class label, and category_id 11 has the lowest count, which is the Title class label). We will need to balance this dataset, most likely using the SMOTE method, to ensure the model sees every class proportionately. The ethical consideration for this dataset is that although the dataset is for

public use, it contains private information such as financial statements and government tenders. Therefore, this dataset will be used for training and testing purposes only and will not be used outside of its intended use (as mentioned in the HuggingFace documentation: https://huggingface.co/datasets/docling-project/DocLayNet).

The PDF Accessibility Benchmark (2025) contains 125 PDF documents aligned with the WCAG 2.2 and PDF /UA standards, which can be accessed through the Github repository (https://github.com/Anukriti12/PDF-Accessibility-Benchmark). There is a total of 7 criteria labels; Alternative text quality, color contrast, font readability, functional hyperlinks, logical reading order, semantic tagging, and table structure. Each PDF document contains images, text, structure information, and metadata to classify into one of the four accessibility labels; Passed, Failed, Not Present, and Cannot Tell. This dataset will be accessed using PyMuPDF from the repository directly to extract all needed PDF content and metadata. However, since this dataset is small, we will need to implement data augmentation to increase the amount in each category. We can use PDFs from the previous dataset and programmatically augment them to contain specific failed criteria (such as Alternative Text). The dataset also contains a "Cannot Tell" category, so some of these PDFs may not be useful to identify a specific criterion. Therefore, these could be excluded from training and only a subset of Pass/Failed PDFs, along with newly created samples, will be given to the model. Some ethical considerations for this dataset is that it can contain private information about differing institutions, authors, and research data. As with the first dataset, proper training use of this dataset is needed while restricting access to any private metadata (only the finalized subset of the data will be needed).

Planned Architecture



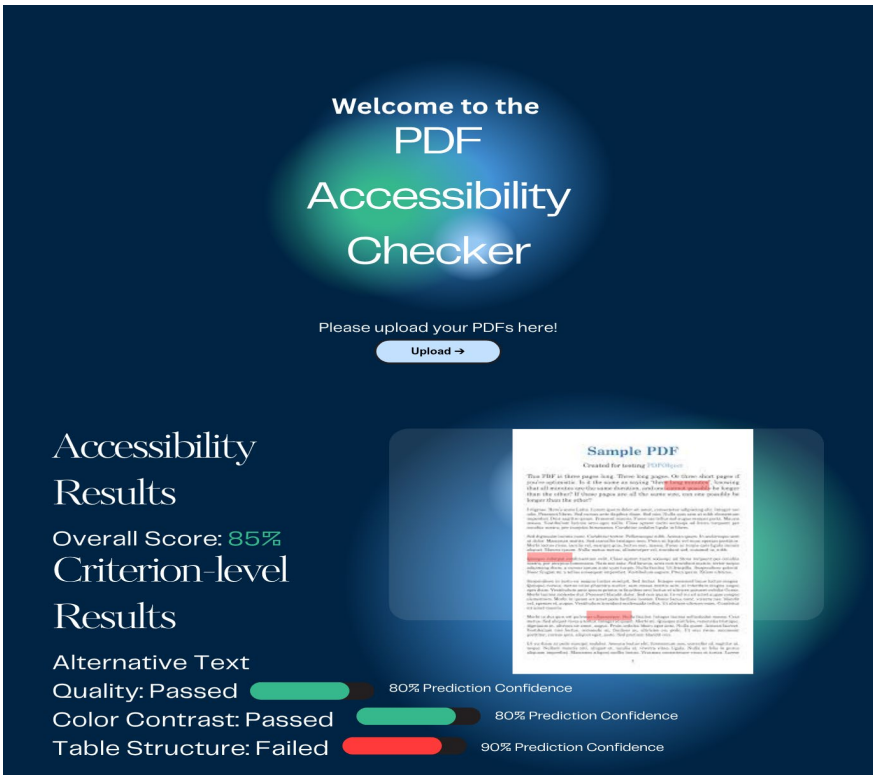
The LayoutLMv3 multimodal transformer model is designed to understand three elements: text, visuals, and spatial information. The PDF pages in both the datasets will need to be converted into images (which both have within their datasets) and the text contained in each PDF, along with the bounding box coordinates used in the DocLayNet dataset, will need to be extracted. These will be the information the model needs to identify the types and locations of the layout elements. It will first run through the first dataset to understand PDF layout elements, and then run through the second dataset to understand accessibility

criteria. We can analyze the results and understand if transfer learning is needed for the multimodal transformer model and if we need a semantic element, such as a CNN model, attached to the current model to understand deeper contextual relationships with graphics.

The user interface will implement Streamlit, since it is very user-friendly and can provide an easily accessible and understandable interface for instructors. The features this interface will contain is a PDF uploader, a preview of the uploaded documents, a results dashboard displaying the accessibility score and associated confidence scores, and an issue identification to indicate where in the document the issue was located. If there is time, other features may be included, such as recommendations for fixing the issues.

User Interface Plan

The user will provide one or multiple PDFs to the interface. The interface will then display a preview of their documents, a results dashboard, and identification of where in the document the violations were found. This will help the instructors understand how much their PDFs have Passed/Failed one of the three criteria and where in the documents they need to be fixed. If they do not need specific criteria indicated, the overall accessibility score can be a summarized report of their PDF(s).



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Innovation and Anticipated Challenges

This model will solve commonly missed errors seen in most public accessibility checkers by focusing on the Alternative Text criteria. Many accidentally pass PDFs if they only include alternative text to a graphic but fail to see if the text connects to the graphic overall. Therefore, the model will be able to check if the semantic meaning behind the alternative text connects with the associated graphic by using multimodal transformer models and experiment with transfer learning. If time is allowed, more pieces to understand contextual information, such as other architectures added on to the current model, can be explored to see if the model is not powerful enough to understand alternative text with graphics. Some risks that can be found will be the data itself, since one dataset is extremely large and the other extremely small. Both will need to be preprocessed carefully, such as dealing with class imbalance, data augmentation, and subsetting data, to only contain valuable information needed for the model. Another difficulty may be the connection with semantic meaning and graphics. Although the model can process text, visual, and spatial information, we do not know how well the model is in classifying the labels in the PDF Accessibility Benchmark dataset. As stated before, other architectures may need to be added on to the model to help in its classification.

Implementation Timeline

Week	Dates	Focus	Outcome
Week 1	Nov 11 – 15	Data Preprocessing and model setup	Clean both datasets and configure LayoutLMv3 model
Week 2	Nov 16 – 22	Model development and training	Implement model through both datasets, calculate confident scores and accessibility scores, record baseline
Week 3	Nov 23 – 29	Refine model and calculations	If needed, add more architectures onto model if training does not show much improvement, recalculate scores if needed, if not, continue with baseline model and retrain until results improve

Week 4	Nov 30 – Dec 6	Streamlit website build and testing	Working PDF upload site, model is working correctly and accurately
Final Days	Dec 7-9	Report and presentation	Project documentation, demonstration, and presentation

Responsible AI Reflection

AI will be used in this project for any repeated processes that need to be calculated, such as helper functions and UI implementation. These will be documented within the coding notebooks to ensure transparency.